Satria Hadi Wiyana

AI Engineer & Machine Learning Engineer | satriahadiwiyana7@gmail.com | linkedin.com/in/satriahadiwiyana

AI/ML Engineer and Technical Project Manager with experience in generative AI, NLP, Computer Vision, RAG/GraphRAG, and full-stack applications. Experienced in leading cross-functional delivery from requirements and architecture through quality assurance and deployment, while developing local model inference for privacy and efficiency. Strong background in technical mentoring, troubleshooting, model evaluation, and translating business requirements into practical AI solutions.

WORK EXPERIENCE

Instruktur Artificial Intelligence -Hitaclass 2026 - Present

Mentor Spesialis AI & ML

- Mentor students through personalized 1-on-1 learning paths covering Python, Data Science, Machine Learning, Deep Learning, Computer Vision, and NLP.
- Guide students in applying AI concepts to real-world cases, from problem formulation and approach selection to model implementation and evaluation.
- Provide hands-on guidance on Neural Networks and practical AI workflows to help students understand both fundamental concepts and real-world implementation.
- Support project implementation and debugging so students can independently apply AI concepts beyond theoretical understanding.

Ketua Tim Development -Program Pengabdian Masyarakat 2024 – 2026

Ketua Tim

- Led a multi-year development program focused on designing and implementing web platforms to digitize operational processes and transaction tracking for partner organizations.
- Developed predictive maintenance integration using IoT sensor data to evaluate maintenance needs and improve operational efficiency.
- Combined web development, IoT, and Machine Learning to provide predictive analytics and operational insights.
- Coordinated technical requirements with partner operational needs to ensure that web, IoT, and Machine Learning development remained aligned with business objectives.

Forum Assisten -AMIKOM Yogyakarta 2023 – 2026

Asisten Praktikum

- Assisted more than 50 students per semester in practical sessions covering Algorithms & Programming, Operating Systems, Computer Networks, and Web Programming.
- Conducted hands-on sessions to strengthen students' understanding of programming logic, debugging, and computational problem-solving.
- Provided guidance on both fundamental programming concepts and practical implementation.
- Designed interactive, project-based learning activities to encourage active participation and problem-solving.

Web Master - The Nganjuk Post (TNP) 2020 – 2023

Web Master

- Managed the operational WordPress-based news portal and maintained server stability while serving more than 2,000 monthly readers.
- Optimized website performance and implemented efficient caching to maintain responsive page loading.
- Maintained platform availability and ensured published content remained accurate, responsive, and consistent.
- Performed maintenance and CMS optimization to support stable publication workflows and reliable reader access.

PROJECTS & RELEVANT EXPERIENCE

Opticargo -Platform Logistik Maritim Agentic GraphRAG 2026 - Present

Project Manager

- Act as Project Manager, leading the delivery of an AI-powered maritime logistics platform from requirements and architecture through implementation readiness.
- Manage PRD development, backlog prioritization, ownership, dependencies, and acceptance criteria across the project.
- Coordinate repository structure and development workflows while maintaining clear technical and product priorities.
- Integrate modular components across the platform, including notification systems, document ingestion, Dockerized services, RAG/vector retrieval, knowledge graphs, and AI agents.
- Define separation of concerns between supply data, RAG/vector retrieval, knowledge graph, and AI agent components to maintain clear architectural boundaries and dependencies.
- Establish technical requirements and implementation standards to ensure integration readiness and consistent quality across components.

TryLab -Sistem Pengujian UI/UX Mobile Sintetis 2026 - Present

AI Engineer

- Designed and developed an AI agent capable of autonomously executing usability tests on mobile applications without manual intervention at each step.
- Implemented a local vision-language model pipeline using a 3.2B-parameter model for understanding Android application interfaces.
- Automated Android application navigation using Android Debug Bridge (ADB) and generated interaction reports and usability findings based on simulated user personas.
- Implemented local VLM inference to improve privacy, reduce dependency on external APIs, and support autonomous agent-based testing.

M-RAG IDX -Asisten Analis Finansial AI Kelas Enterprise 2026

AI Engineer

- Designed a Multimodal Retrieval-Augmented Generation (RAG) architecture for retrieving and analyzing annual reports from IDX-listed companies.
- Implemented hybrid retrieval combining semantic relevance with financial data such as net income, NPL, and NPM.
- Developed guardrails and audit trails to map generated answers back to source pages for improved traceability and reliability.
- Designed multimodal data flow integrating text, tables, financial metrics, and visual information into a unified analytical pipeline.

PANGAN-ALERT - Explainable Spatio-Temporal Early Warning 2026

Data Scientist

- Developed PANGAN-ALERT, an early warning system designed to predict food security index declines for the following year across 514 districts/cities in Indonesia.
- Implemented supervised learning using spatial and temporal data, including CHIRPS precipitation, DHI, and temporal split strategies based on KNN-5.
- Prepared a supervised panel dataset containing 3,084 observations with anti-leakage feature selection and temporal split policies to maintain evaluation validity.

Sustainability Resilience Intelligence -Rantai Pasok Benih Padi Bali 2026

Data Scientist

- Implemented a feature interaction approach using Resilience Score and an Optimizer to support model analysis.
- Applied Random Forest and SHAP for feature importance analysis to identify attributes with the greatest influence on the sustainability index, achieving an exploratory R^2 of approximately 0.80.
- Combined CNN-based features, SHAP, and actor-gap majority concepts to analyze sustainability-related visual and contextual information.

CERTIFICATIONS & TRAINING

DBS Foundation x Dicoding Coding Camp
2026

Scholarship Recipient - AI Engineer Learning Path

Intensive industry-standard training focused on Machine Learning architecture design, training, and optimization, as well as Deep Learning, supported by hands-on implementation through real-world case studies to evaluate model efficiency and scalability for AI deployment.

FIT 2026 Data Science Competition
2026

Honorable Mention 2 - 2nd International Rank

International data science competition focused on complex dataset analysis, predictive Machine Learning model development, and data-driven solution development to address real-world challenges.

LM-Based Tools and Gemini API Integration for Data Scientists
2026

Training - Hacktiv8 Indonesia

Training focused on Large Language Models and Gemini API integration into Data Science workflows.

Membangun Proyek Deep Learning Tingkat Mahir
2026

Training - Dicoding Indonesia

Advanced training in Deep Learning project development and model implementation through practical case studies.

Spec-Driven Development dengan Kiro
2026

Training - Dicoding Indonesia

Training focused on software development using specification-driven workflows.

Excel Intermediate
2024

Training - Coding Studio

Intermediate-level training in Microsoft Excel for data processing and analysis.

PENDIDIKAN

Universitas Amikom - Yogyakarta
2023 - Present

Bachelor of Informatics (S1) , GPA: 3.76/4.00

Focus areas include Artificial Intelligence, Machine Learning, Data Science, and Software Engineering, with practical application through AI projects, data analysis, and software development.

TECHNICAL EXPERTISE

- **AI/ML & Generative AI** : LLM, RAG/GraphRAG, VLM, NLP, Computer Vision, Deep Learning, local inference, TensorFlow, Keras, Scikit-learn, XGBoost, CatBoost, Random Forest.
- **Data Science & XAI** : data cleaning, EDA, feature engineering, model evaluation, SHAP/XAI, spatio-temporal analysis, Pandas, NumPy, Matplotlib, Seaborn, dan Microsoft Excel (Intermediate).
- **Data & Geospatial Engineering** : PySpark, Hadoop, GeoPandas, Google Earth Engine, ETL/tidy-data pipeline, serta geospatial/climate-data processing.
- **Software Engineering** : Python, JavaScript, React.js, Laravel, Flask, Streamlit, REST API, Git, Docker, dan ADB.
- **Databases & Retrieval** : PostgreSQL, MySQL, Redis, MinIO, Neo4j, Qdrant, hybrid retrieval, dan vector search.